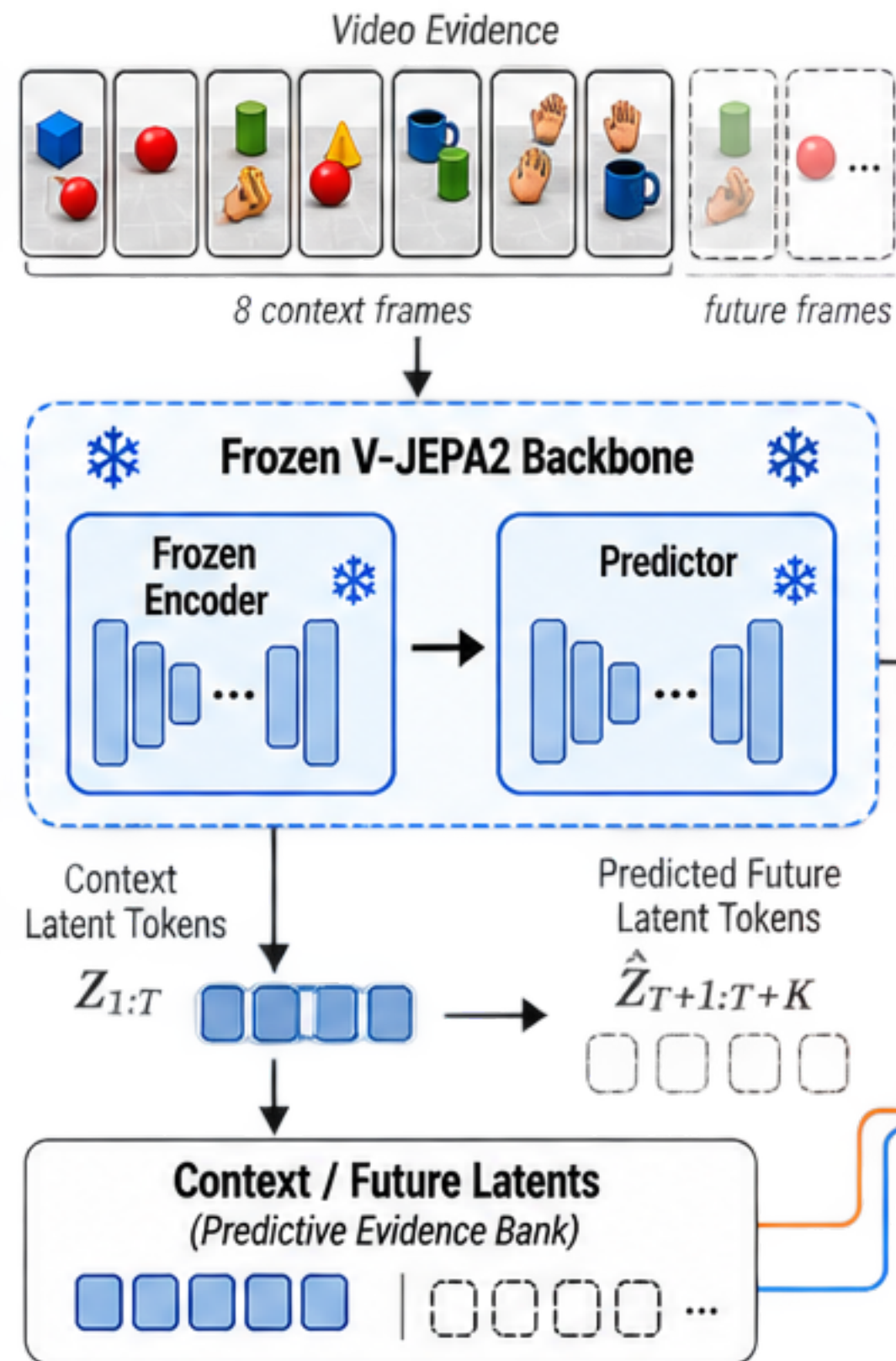


1) Input Evidence and Frozen Backbone



$$X \rightarrow (O, D, R) \rightarrow Y$$

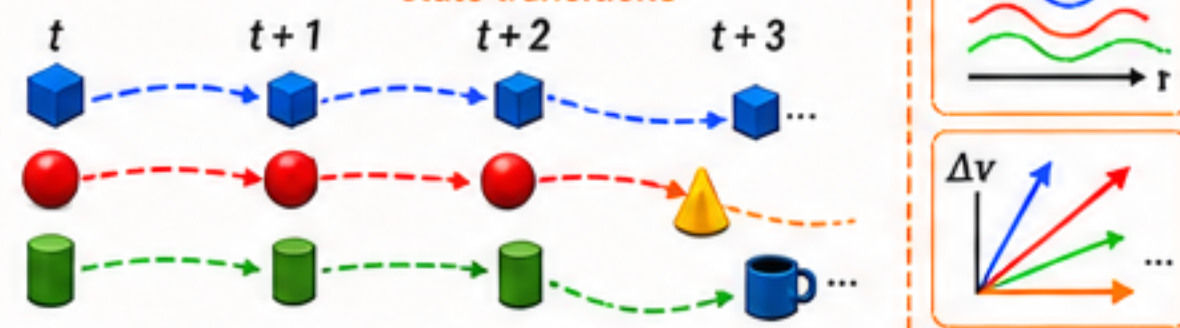
3) Relation State

Pair Construction → Interaction Graph
pairwise interactions



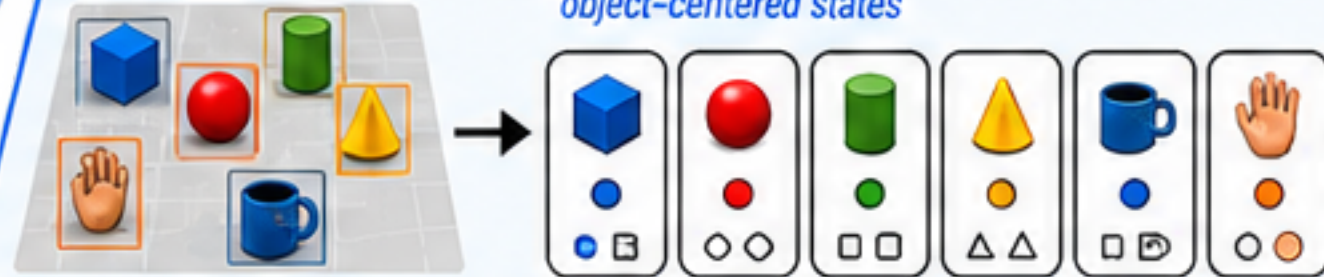
2) Dynamic State

Entity Trajectories → State Transitions
state transitions



1) Entity State

Patch Evidence → Object Slots
object-centered states



0) Context / Future Latents

visual token sequences from frozen backbone

3) Unified Output Paths

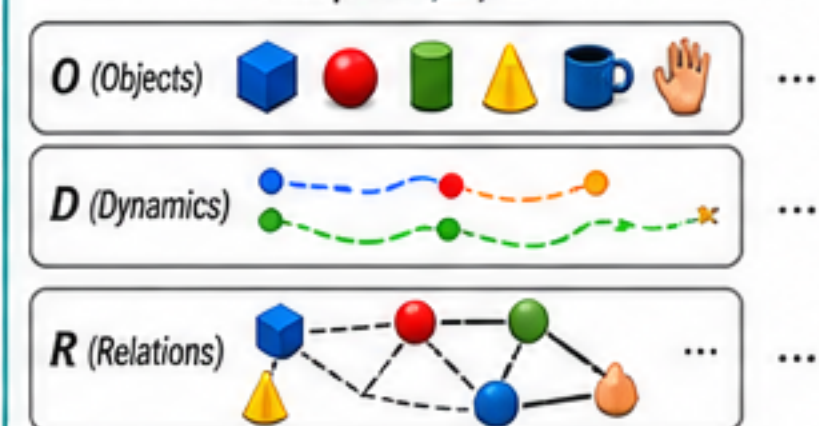
Prediction

Structured State $S^* \rightarrow$ Future Frames / Events



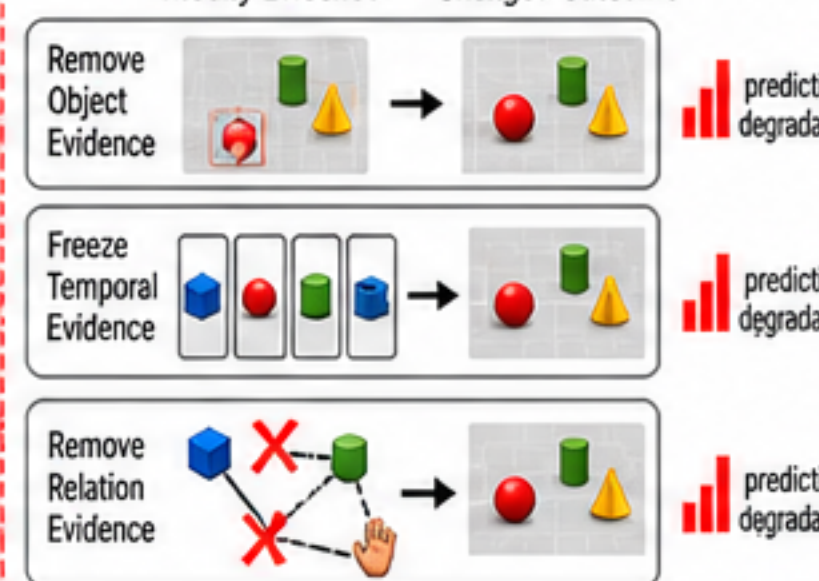
Explanation / Decode

interpret O, D, R



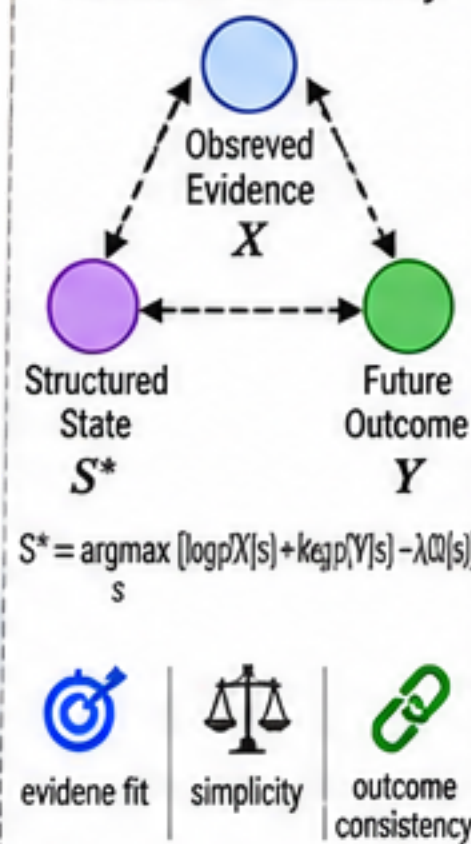
Intervention

Modify Evidence → Changed Outcome



Structured State
 $S = (O, D, R)$

Abductive Consistency



Shared Structured State Interface

